

Second Chapter Title

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1 Introduction

This chapter develops the main technical apparatus. Building on the Kripke semantics of Chapter ??, we discuss frame conditions and introduce neighborhood semantics.

2 Frame Conditions

Properties of the accessibility relation correspond to axiom schemas.

Proposition 2.1. *A frame $\mathfrak{F}_R$ validates $\mathbf{T}$ ($\Box A \rightarrow A$) if and only if R is reflexive.*

Proof. (LTR) Fix $w \in W$ and set $V(p) = \{v : wRv\}$. Let $\mathfrak{M} = \langle W, R, V \rangle$. Then $\mathfrak{M}, w \Vdash \Box p$, so $\mathbf{T}$ gives $\mathfrak{M}, w \Vdash p$, hence $w \in V(p)$, i.e., wRw .

(RTL) If R is reflexive and $\mathfrak{M}, w \Vdash \Box A$, then wRw gives $\mathfrak{M}, w \Vdash A$. $\square$

Note 2.2. Transitivity corresponds to $\Box A \rightarrow \Box \Box A$ and symmetry to $A \rightarrow \Box \Diamond A$. See Blackburn, Rijke, and Venema [1, ch. 3].

3 Neighborhood Semantics

Neighborhood semantics generalize relational semantics by replacing R with a set-selection function [4, 3].

Definition 3.1. A *neighborhood model* $\mathfrak{M}_N = \langle W, f_N, V \rangle$ has $f_N : W \rightarrow \wp(\wp(W))$ assigning each world a collection of neighborhoods. The modal clause is:

$$\mathfrak{M}_N, w \Vdash \Box A \text{ if and only if } \llbracket A \rrbracket^{\mathfrak{M}_N} \in f_N(w).$$

Remark 3.2. Every Kripke frame induces a neighborhood frame via $f_N(w) = \{X \subseteq W : \{v : wRv\} \subseteq X\}$. The converse fails: neighborhood frames can model non-normal logics lacking $\mathbf{K}$ or $\mathbf{Nec}$ [2].

4 Supplementation

Definition 4.1. f_N is *supplemented* if $X \in f_N(w)$ and $X \subseteq Y$ imply $Y \in f_N(w)$ for all w .

Lemma 4.2. *Every f_N has a unique smallest supplemented extension f_N^+ given by $f_N^+(w) = \{Y : X \subseteq Y \text{ for some } X \in f_N(w)\}$.*

Proof. $f_N(w) \subseteq f_N^+(w)$ is immediate. If $Y \in f_N^+(w)$ and $Y \subseteq Z$, the witnessing X gives $X \subseteq Z$, so $Z \in f_N^+(w)$. Any supplemented extension must contain each such Y , giving minimality. $\square$

References

- [1] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Vol. 53. Cambridge Tracts in Theoretical Computer Science. Cambridge: Cambridge University Press, 2001.
- [2] Brian F. Chellas. *Modal Logic: An Introduction*. Cambridge: Cambridge University Press, 1980.
- [3] Eric Pacuit. *Neighborhood Semantics for Modal Logic*. Cham, Switzerland: Springer, 2017.
- [4] Dana Scott. “Advice on Modal Logic”. In: *Philosophical Problems in Logic*. Ed. by Karel Lambert. Vol. 29. Synthese Library. Dordrecht: Springer, 1970, pp. 143–173.